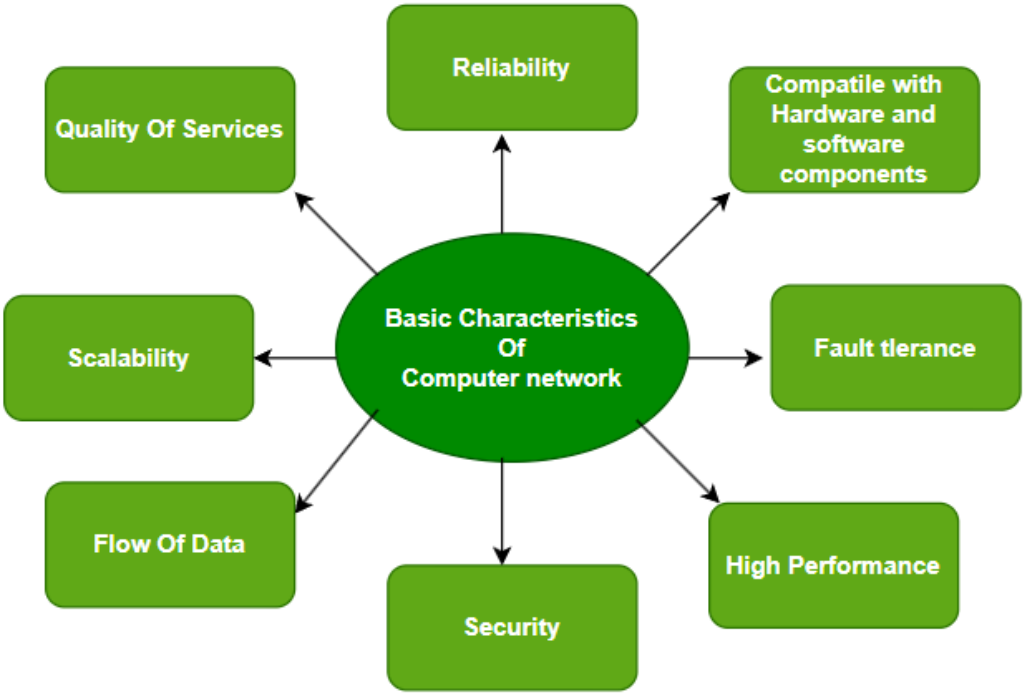
Computer Network

Unit-I: Introduction to Data Communication and Computer Network

❖ Data Communication

Definition: is defined as a process in which more than one computer transfers information, instructions to each other and for sharing resources.

❖ Basic Characteristics of Computer Network:



characteristics of Computer Networks

1. Security

Security is one of the most important characteristics of a computer network. Most businesses nowadays depend on computers, which are accessed by using network. If computer network technology is not secure, then any one can able to access companies data and this is not good for any company, because those who unauthorized data can misuse of the data. However, nowadays, computer networking tools primarily provide the highest level of security and prevent any unauthorized access.

Computer networks are vulnerable to security threats such as hacking, viruses, and data breaches. Security measures such as firewalls, encryption, and user authentication are essential to protect network resources and data.

2. Reliability

Computer networks must be reliable to ensure that data and resources are always available when needed. Redundancy and backup systems can help to ensure that the network remains operational in the event of a failure.

3. Scalability

Scalability means a network can grow and handle more users or devices without losing performance. The internet is a great example of this: even as millions of new users connect and communicate with other devices, the network continues to work well.

Computer networks must be scalable to accommodate growth and changing needs. As the number of devices and users on the network increases, the network must be able to handle the additional traffic and data.

4. Flow of Data

Computer networks let users access and share data like files and documents between devices. This feature is essential because it enables data to move smoothly from one device to another.

5. High Performance

Performance is measured by how quickly a command is executed. If data transfers fast and responses are quick, it benefits users by making data sharing and resource usage more efficient. Using multiple processors can further enhance performance.

The performance of a computer network is determined by factors such as bandwidth, latency, and throughput. These factors affect the speed and responsiveness of the network and can impact the user experience.

6. Fault Tolerance

Fault tolerance is a valuable feature of computer networks. For example, if two devices are connected by both wired and wireless means, and the wireless connection fails, the data can still be sent through the wired connection. This ensures that communication continues smoothly even if part of the network is down or damaged.

7. Quality of Service (QoS)

It means people can choose what data gets sent first and how it's sent, making sure it goes fast. If some data doesn't make it through, the system can handle that too. This makes sure users get a good experience when using the network.

8. Compatible with Hardware and Software Components

Another important characteristics of computer network is that it lets lots of devices use the same software. This means you can use the same program on different hardware. This makes things work better together and makes software easier to use. Plus, it helps make the most out of the stuff you have.

- **Characteristics of Data Communication System**

Data communication is the exchange of data between two devices by using transmission medium (Wire or Wireless).

The effectiveness of a data communications system depends on some fundamental factors:

1) Delivery 2) Accuracy 3) Timeliness 4) Jitter 5) Error detection & Correction

Delivery:

System must deliver the data to the correct destination. (MAC address, IP Address)

Accuracy:

The system must deliver the data accurately. Altered data is unusable.

Timeliness:

In the case of video and audio, late delivery is useless. Communication system must deliver data in a timely manner. Timely means delivering data without any delay (real-time transmission).

Jitter:

Variation in the packet arrival time. It is the delay in the delivery of audio or video data packets. an uneven quality in the video is the result.

Error Detection & Correction:

Parity Check

Checksum

CRC

❖ Components of Computer Network

The physical components of a computer network include hardware devices and media that enable connectivity and data exchange between devices. The server, client, peer, transmission media, and connecting devices make up the hardware components. The operating system and protocols are examples of software components. A computer network is made up of several computers connected so that resources and data can be shared. In this article, we will discuss every point about the physical component of a computer network.

❖ Types of Physical Components

1. NIC (Network Interface Card):

NIC or Network Interface Card is a network adapter used to connect the computer to the network. It is installed in the computer to establish a LAN. It has a unique ID that is written on the chip, and it has a connector to connect the cable to it. The cable acts as an interface between the computer and the router or modem. NIC card is a layer 2 device, which means it works on the network model's physical and data link layers.

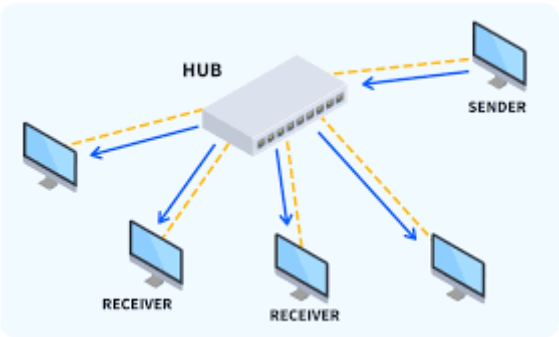
Types of NIC

Wired NIC: Cables and Connectors use Wired NIC to transfer data.

Wireless NIC: These connect to a wireless network such as WiFi, Bluetooth, etc.

2. HUB

A hub is a multi-port repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices. In other words, the collision domain of all hosts connected through hub remains one. Hub does not have any routing table to store the data of ports and map destination addresses., the routing table is used to send/broadcast information across all the ports.



Types of HUB

Active HUB: Active HUB regenerates and amplifies the electric signal before sending them to all connected device. This hub is suitable to transmit data for long distance connections over the network.

Passive HUB: As the name suggests it does not amplify or regenerate electric signal, it is the simplest types of Hub among all and it is not suitable for long-distnace connections.

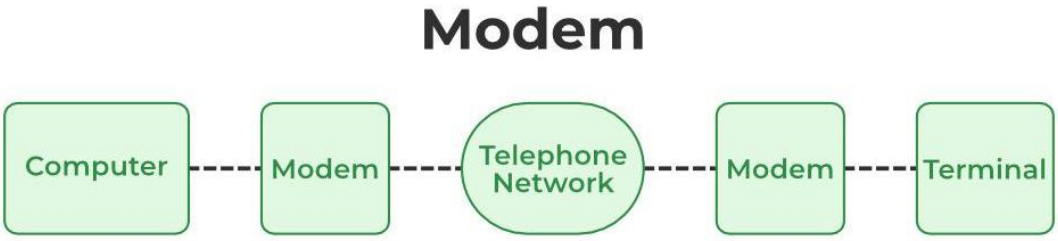
Switching HUB: This is also known as intelligent HUB, they provide some additional functionality over active and passive hubs. They analyze data packets and make decisions based on MAC address and they are operated on DLL (Data Link Layer).

3. Router

A Router is a device like a switch that routes data packets based on their IP addresses. The router is mainly a Network Layer device. Routers normally connect LANs and WANs and have a dynamically updating routing table based on which they make decisions on routing the data packets. The router divides the broadcast domains of hosts connected through it.

4. Modem

A Modem is a short form of Modulator/Demodulator. The Modem is a hardware component/device that can connect computers and other devices such as routers and switches to the internet. Modems convert or modulate the analog signals coming from telephone wire into a digital form that is in the form of 0s and 1s.



5. Switch

A Switch is a multiport bridge with a buffer and a design that can boost its efficiency (a large number of ports implies less traffic) and performance. A switch is a data link layer device. The switch can perform error checking before forwarding data, which makes it very efficient as it does not forward packets that have errors and forward good packets selectively to the correct port only.

6. Nodes

Node is a term used to refer to any computing devices such as computers that send and receive network packets across the network.

7. Media

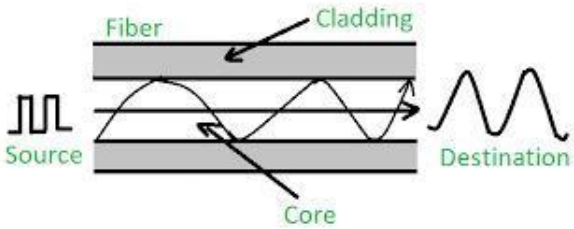
It is also known as Link which is going to carry data from one side to another side.

This link can be Wired Medium (Guided Medium) and Wireless Medium (Unguided Medium).

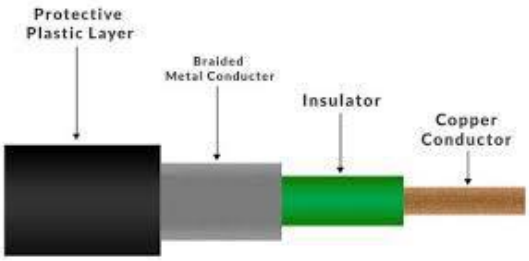
7.1 Wired Medium

Ethernet: Ethernet is the most widely used LAN technology, which is defined under IEEE standards 802.3.

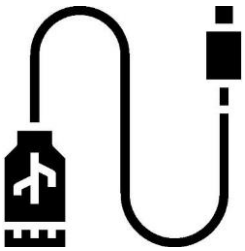
Fibre Optic Cable: In fibre optic cable data is transferred in the form of light waves.



Coaxial Cable: Coaxial Cable is mainly used for audio and video communications.



USB Cable: USB Stands for Universal Serial Bus it is mainly used to connect PCs and smartphones.



7.2 Wireless Medium

Infrared (E.g. short-range communication – TV remote control).

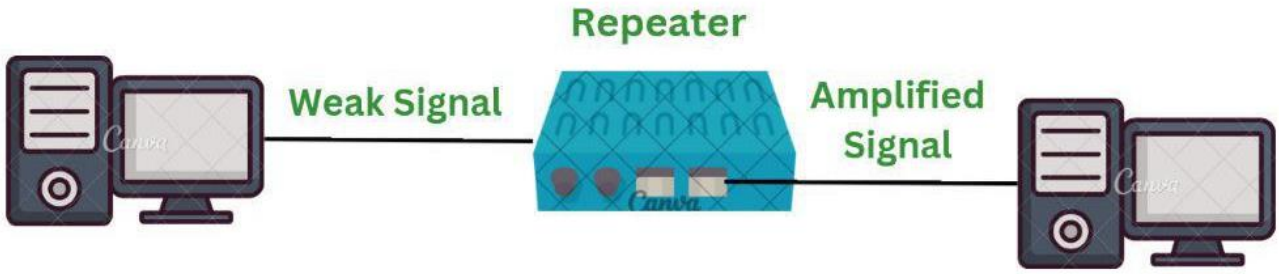
Radio (E.g. Bluetooth, Wi-Fi).

Microwaves (E.g. Cellular system).

Satellite (E.g. Long range communications – GPS).

8. Repeater

Repeater is an important component of computer networks as it is used to regenerate and amplify signal in the computer networks. Repeaters are used to improve the quality of the networks and they are operated on the Physical Layer of the OSI Model.



9. Server

A server is a computer program that provides various functionality to another computer program. The server plays a vital role in facilitating communication, data storage, etc. Servers have more data storage as compared to normal computers. They are designed for the specific purpose of handling multiple requests from clients.

❖ Need of Computer Network

Computer networks have become vital in the modern age because they allow interaction as well as sharing of resources among various devices and individuals. The following is a list of things that we do via computer networks, matters that we benefit from using computer networks, or items that have become achievable or in effect as a result of computer networks. Let us know all the details so that we can understand the significance of computer networking.

- **Easy resource sharing** – Computer networks allow users and devices to share resources like printers, documents, and apps easily. This increases efficiency while decreasing the expense of offering resources.
- **Good communication facility** – Computer networks help individuals to communicate with one another regardless of where they are. Individuals can communicate in real-time using networking tools such as instant messaging, email, and video conferencing.
- **Access information** – Networks enable access to a diverse set of services and information, such as the internet, database servers, and cloud-based services for computing. Individuals can now obtain and share data all over the globe.
- **High security** – Networks enable the deployment of security precautions like encrypted connections, firewalls, and access control that help protect sensitive information and data from illicit access and other safety hazards.
- **Collaboration** – Networks enable collaboration among people and groups engaged in the same task, regardless of their geographical location. This encourages collaboration, improves productivity, and fosters creativity.
- **Centralized management** – Networks enable centralized handling of resources, apps, and information.

❖ Applications of Computer Networking

Some of the applications of Computer Networking are discussed below:

- **Resource Sharing:** Resource Sharing is one of the important applications of Computer Networking. You can share a single software among Multiple users. We can also share Hardware Devices via this technique.
- **Communication:** Communication Medium means various ways through which we can communicate like Email Calls, broadcasts, etc.
- **Home Applications:** Home Applications are an important application of Computer Networking. Examples are User-to-User Communication, Entertainment, E-Commerce, etc.
- **Business Applications:** Business Application is also an important application of Computer Networking. Almost all companies are doing business online with the help of Computer Networking.
- **Social Media:** Social Media is one of the recent and widely used applications of Computer Networking. It helps people in getting news, feed, current trending topics of all types, etc.
- **Access to Remote Information:** Computer Networking helps in accessing remote information from the end-users.

  For example: details required for train tickets, plane tickets, etc.

- **Cloud Computing:** Computer Networking has facilitated the development of cloud computing, which allows users to store and access data and applications from remote servers via the internet.
- **Virtualization:** Networking technologies also enable virtualization, which allows multiple virtual machines to run on a single physical machine, improving resource utilization and reducing costs.
- **Online Education:** Computer Networking has revolutionized education by providing online learning platforms, distance education, and virtual classrooms.
- **Remote Working:** Networking technologies have enabled remote working, allowing employees to work from home or other locations outside of the office. This has become especially important during the COVID-19 pandemic.
- **E-commerce:** Computer Networking has facilitated the growth of e-commerce by allowing businesses to sell their products and services online and reach a global market.
- **Telemedicine:** Networking technologies have also enabled telemedicine, allowing doctors and healthcare providers to provide remote medical consultations and diagnosis to patients in remote locations.
- **Real-time Collaboration:** Networking technologies enable real-time collaboration, allowing people to work together on projects and share information in real-time, no matter where they are located.

❖ **Goals of Computer Networks:**

The following are some important goals of computer networks:

- **Resource Sharing:** Many organization has a substantial number of computers in operations, which are located apart. Ex. A group of office workers can share a common printer, fax, modem, scanner, etc.
- **High Reliability:** If there are alternate sources of supply, all files could be replicated on two or more machines. If one of them is not available, due to hardware failure, the other copies could be used.
- **Inter-Process Communication:** Network users, located geographically apart, may converse in an interactive session through the network. In order to permit this, the network must provide almost error-free communications.
- **Flexible Access:** Files can be accessed from any computer in the network. The project can be begun on one computer and finished on another.
- **Security**: Computer networks must be secure to protect against unauthorized access, data breaches, and other security threats. This includes implementing measures such as firewalls, antivirus software, and encryption to ensure the confidentiality, integrity, and availability of data.
- **Performance**: Computer networks must provide high performance and low latency to ensure that applications and services are responsive and available when needed. This requires optimizing network infrastructure, bandwidth utilization, and traffic management.
- **Scalability:** Computer networks must be designed to scale up or down as needed to accommodate changes in the number of users, devices, and data traffic. This requires careful planning and management to ensure the network can meet current and future needs.

❖ Advantages of Computer Networking

Some of the main advantages of Computer Networking are discussed below:

- **Central Storage of Data:** Files can be stored on a central node (the file server) that can be shared and made available to every user in an organization.
- **Connectivity:** There is a negligible range of abilities required to connect to a modern computer network. The effortlessness of joining makes it workable for even youthful kids to start exploring the data.
- **Faster Problem-solving:** Since an extensive procedure is disintegrated into a few littler procedures and each is taken care of by all the associated gadgets, an explicit issue can be settled in lesser time.
- **Reliability:** Reliability implies backing up information. Due to some reason equipment crashes, and so on, the information gets undermined or inaccessible on one PC, and another duplicate of similar information is accessible on another workstation for future use, which prompts smooth working and further handling without interruption.
- **Flexible:** This innovation is known to be truly adaptable, as it offers clients the chance to investigate everything about fundamental things, for example, programming without influencing their usefulness.
- **Security through Authorization:** Security and protection of information are additionally settled through the system. As just the system clients are approved to get to specific records or applications, no other individual can crack the protection or security of information.
- **Storage capacity:** Since you will share data, records, and assets with other individuals, you need to guarantee all information and substance are legitimately put away in the framework. With this systems administration innovation, you can do most of this with no issue, while having all the space you require for capacity.

❖ Disadvantages of Computer Networking

Some of the main disadvantages of Computer Networking are discussed below:

- **Expensive:** Execution of the network can be expensive in the case of an initial setup, as the wires and the cost of the cable are high and sometimes equipment is also costly.
- **Virus and Malware:** Computer Networking can lead to the spreading of viruses to another computer through the network.
- **Management of the network:** Management of the Network is quite difficult as it requires skilled persons to handle that large network. It requires training of people who are employed in this work.
- **Loss of Information:** In case of a crash of the Computer Network, it can lead to the loss of information or not being able to access information for some time.
- **The system can be Hacked:** In the case of Wide Area Networks(WAN), there is a threat of Hacking of the System. Some security features should be added to prevent such things.

Types of Network Topology

- Network topology refers to the arrangement of different elements like nodes, links, and devices in a computer network. It defines how these components are connected and interact with each other. Understanding various types of network topologies helps in designing efficient and robust networks. Common types include bus, star, ring, mesh, and tree topologies, each with its own advantages and disadvantages. In this article, we are going to discuss different types of network topology their advantages and disadvantages in detail.

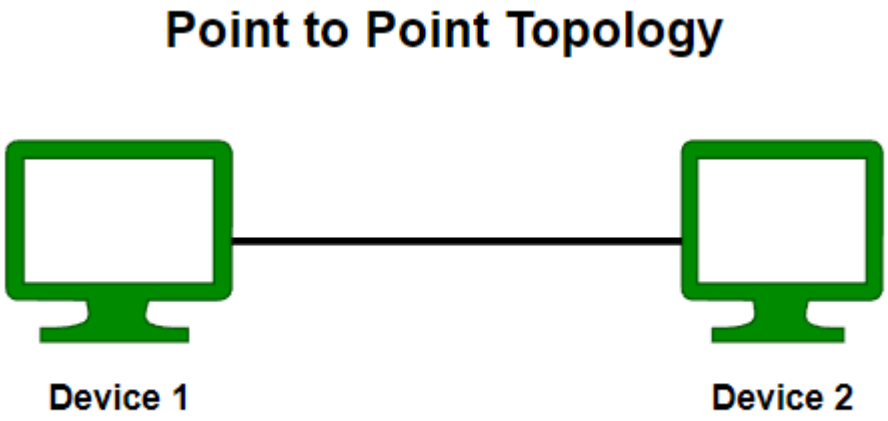
Types of Network Topology

The arrangement of a network that comprises nodes and connecting lines via sender and receiver is referred to as **Network Topology**. The various network topologies are:

- Point to Point Topology
- Mesh Topology
- Star Topology
- Bus Topology
- Ring Topology
- Tree Topology
- Hybrid Topology

Point to Point Topology

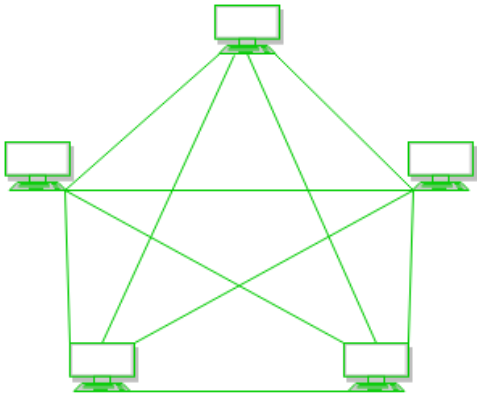
Point-to-point topology is a type of topology that works on the functionality of the sender and receiver. It is the simplest communication between two nodes, in which one is the sender and the other one is the receiver. Point-to-Point provides high bandwidth.



Point to Point Topology

Mesh Topology

In a mesh topology, every device is connected to another device via a particular channel. In Mesh Topology, the protocols used are AHCP (Ad Hoc Configuration Protocols), DHCP (Dynamic Host Configuration Protocol), etc.



Mesh Topology

Advantages of Mesh Topology

- Communication is very fast between the nodes.
- Mesh Topology is robust.
- The fault is diagnosed easily. Data is reliable because data is transferred among the devices through dedicated channels or links.
- Provides security and privacy.

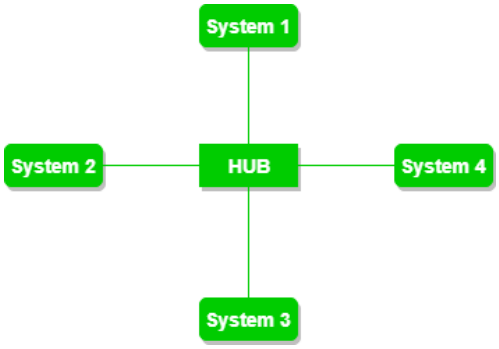
Disadvantages of Mesh Topology

- Installation and configuration are difficult.
- The cost of cables is high as bulk wiring is required, hence suitable for less number of devices.
- The cost of maintenance is high.

A common example of mesh topology is the internet backbone, where various internet service providers are connected to each other via dedicated channels. This topology is also used in military communication systems and aircraft navigation systems.

Star Topology

In Star Topology, all the devices are connected to a single hub through a cable. This hub is the central node and all other nodes are connected to the central node. The hub can be passive in nature i.e., not an intelligent hub such as broadcasting devices, at the same time the hub can be intelligent known as an active hub. Active hubs have repeaters in them. Coaxial cables or RJ-45 cables are used to connect the computers. In Star Topology, many popular Ethernet LAN protocols are used as CD(Collision Detection), CSMA (Carrier Sense Multiple Access), etc.



Star Topology

Advantages of Star Topology

- If N devices are connected to each other in a star topology, then the number of cables required to connect them is N. So, it is easy to set up.
- Each device requires only 1 port i.e. to connect to the hub, therefore the total number of ports required is N.
- It is Robust. If one link fails only that link will affect and not other than that.
- Easy to fault identification and fault isolation.
- Star topology is cost-effective as it uses inexpensive coaxial cable.

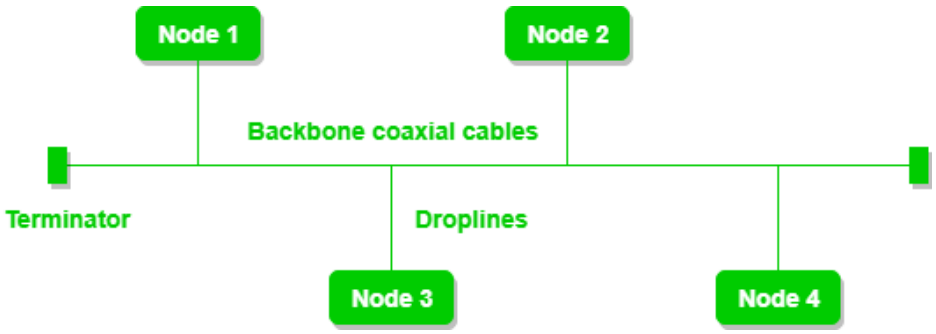
Disadvantages of Star Topology

- If the concentrator (hub) on which the whole topology relies fails, the whole system will crash down.
- The cost of installation is high.
- Performance is based on the single concentrator i.e. hub.

A common example of star topology is a local area network (LAN) in an office where all computers are connected to a central hub. This topology is also used in wireless networks where all devices are connected to a wireless access point.

Bus Topology

Bus Topology is a network type in which every computer and network device is connected to a single cable. It is bi-directional. It is a multi-point connection and a non-robust topology because if the backbone fails the topology crashes. In Bus Topology, various MAC (Media Access Control) protocols are followed by LAN ethernet connections like TDMA, Pure Aloha, CDMA, Slotted Aloha, etc.



Bus Topology

Advantages of Bus Topology

- If N devices are connected to each other in a bus topology, then the number of cables required to connect them is 1, known as backbone cable, and N drop lines are required.
- Coaxial or twisted pair cables are mainly used in bus-based networks that support up to 10 Mbps.
- The cost of the cable is less compared to other topologies, but it is used to build small networks.
- Bus topology is familiar technology as installation and troubleshooting techniques are well known.
- CSMA is the most common method for this type of topology.

Disadvantages of Bus Topology

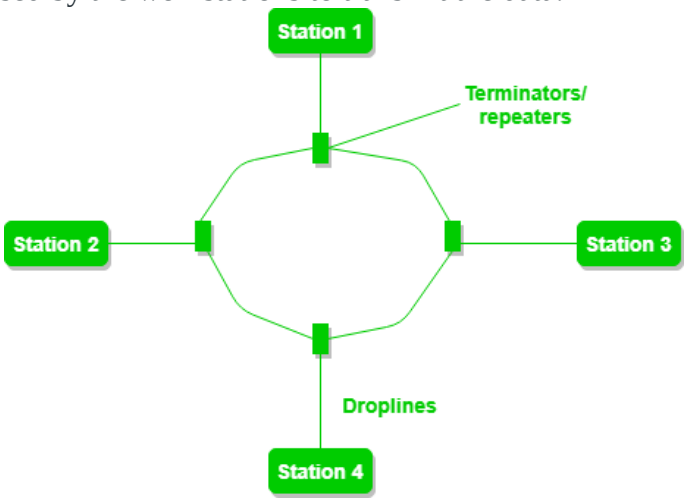
- A bus topology is quite simpler, but still, it requires a lot of cabling.
- If the common cable fails, then the whole system will crash down.
- If the network traffic is heavy, it increases collisions in the network. To avoid this, various protocols are used in the MAC layer known as Pure Aloha, Slotted Aloha, CSMA/CD, etc.
- Adding new devices to the network would slow down networks.
- Security is very low.

A common example of bus topology is the Ethernet LAN, where all devices are connected to a single coaxial cable or twisted pair cable. This topology is also used in cable television networks.

Ring Topology

In a Ring Topology, it forms a ring connecting devices with exactly two neighbouring devices. A number of repeaters are used for Ring topology with a large number of nodes, because if someone wants to send some data to the last node in the ring topology with 100 nodes, then the data will have to pass through 99 nodes to reach the 100th node. Hence to prevent data loss repeaters are used in the network.

The data flows in one direction, i.e. it is unidirectional, but it can be made bidirectional by having 2 connections between each Network Node, it is called Dual Ring Topology. In-Ring Topology, the Token Ring Passing protocol is used by the workstations to transmit the data.



Ring Topology

Advantages of Ring Topology

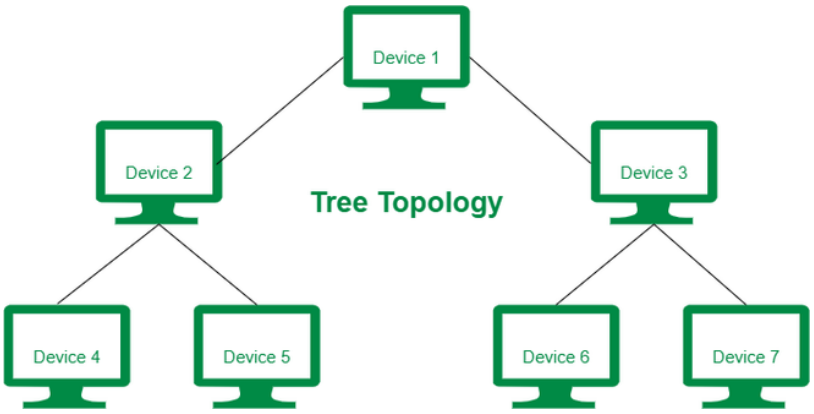
- The data transmission is high-speed.
- The possibility of collision is minimum in this type of topology.
- Cheap to install and expand.
- It is less costly than a star topology.

Disadvantages of Ring Topology

- The failure of a single node in the network can cause the entire network to fail.
- Troubleshooting is difficult in this topology.
- The addition of stations in between or the removal of stations can disturb the whole topology.
- Less secure.

Tree Topology

This topology is the variation of the Star topology. This topology has a hierarchical flow of data. In Tree Topology, protocols like DHCP and SAC (Standard Automatic Configuration) are used.



Tree Topology

Advantages of Tree Topology

- It allows more devices to be attached to a single central hub thus it decreases the distance that is travelled by the signal to come to the devices.
- It allows the network to get isolated and also prioritize from different computers.
- We can add new devices to the existing network**.**
- Error detection and error correction are very easy in a tree topology.

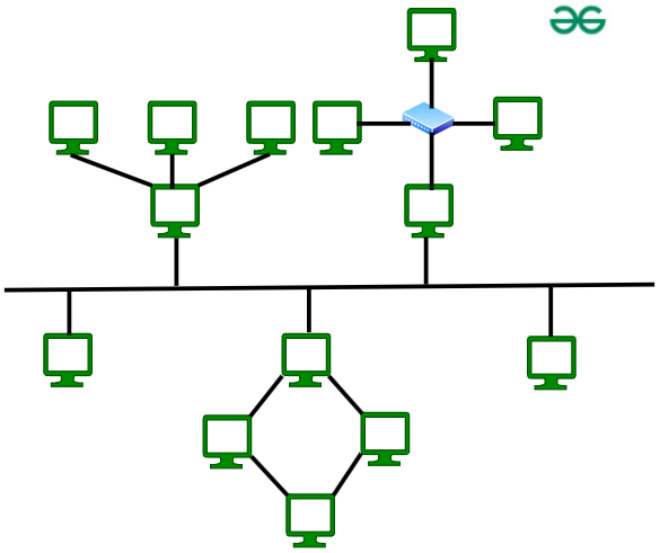
Disadvantages of Tree Topology

- If the central hub gets fails, the entire system fails.
- The cost is high because of the cabling.
- If new devices are added, it becomes difficult to reconfigure.

A common example of a tree topology is the hierarchy in a large organization. At the top of the tree is the CEO, who is connected to the different departments or divisions (child nodes) of the company. Each department has its own hierarchy, with managers overseeing different teams (grandchild nodes). The team members (leaf nodes) are at the bottom of the hierarchy, connected to their respective managers and departments.

Hybrid Topology

This topological technology is the combination of all the various types of topologies we have studied above. Hybrid Topology is used when the nodes are free to take any form. It means these can be individuals such as Ring or Star topology or can be a combination of various types of topologies seen above. Each individual topology uses the protocol that has been discussed earlier.



Hybrid Topology

Advantages of Hybrid Topology

- This topology is **very flexible**.
- The size of the network can be easily expanded by **adding new devices.**

Disadvantages of Hybrid Topology

- It is challenging **to design the architecture** of the Hybrid Network.
- **Hubs** used in this topology are **very expensive.**
- The infrastructure cost is very high as a hybrid network **requires a lot of cabling and network devices**.

A common example of a hybrid topology is a university campus network. The network may have a backbone of a star topology, with each building connected to the backbone through a switch or router. Within each building, there may be a bus or ring topology connecting the different rooms and offices. The wireless access points also create a mesh topology for wireless devices. This hybrid topology allows for efficient communication between different buildings while providing flexibility and redundancy within each building.

Types of Computer Network

A computer network is a cluster of computers over a shared communication path that works to share resources from one computer to another, provided by or located on the network nodes. In this article, we will discuss computer networks and their types.

What is a Computer Network?

A computer network is a system that connects many independent computers to share information (data) and resources. The integration of computers and other different devices allows users to communicate more easily. A computer network is a collection of two or more computer systems that are linked together. A network connection can be established using either cable or wireless media. Hardware and software are used to connect computers and tools in any network.

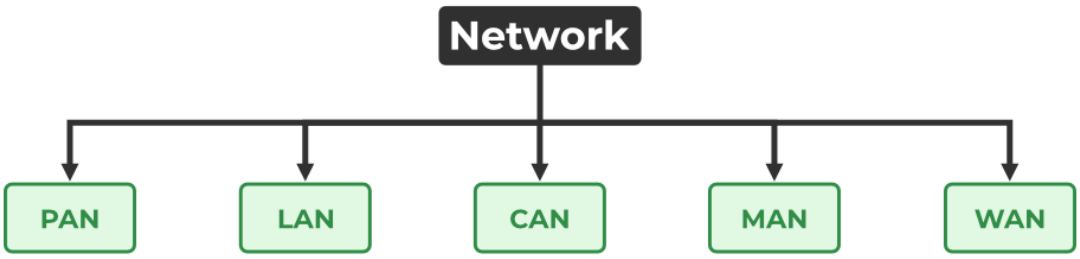
Uses of Computer Networks

- Communicating using email, video, instant messaging, etc.
- Sharing devices such as printers, scanners, etc.
- Sharing files.
- Sharing software and operating programs on remote systems.
- Allowing network users to easily access and maintain information.

Types of Computer Networks

There are mainly five types of Computer Networks

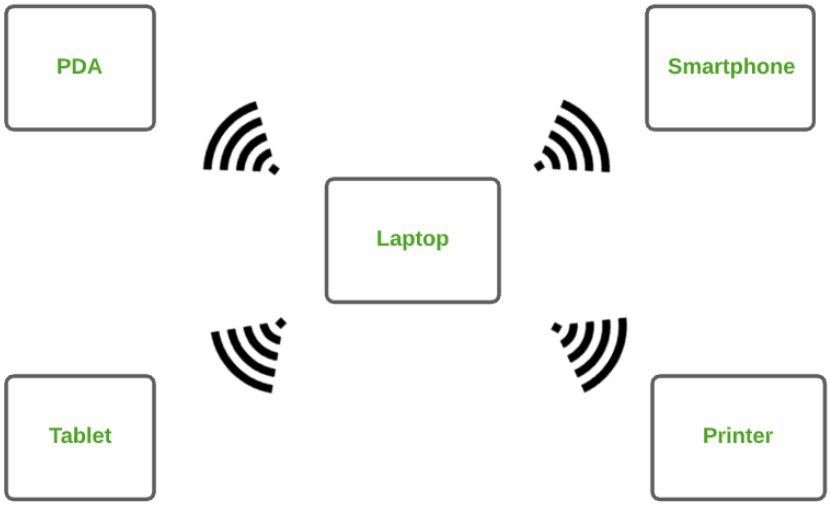
1. Personal Area Network (PAN)
2. Local Area Network (LAN)
3. Campus Area Network (CAN)
4. Metropolitan Area Network (MAN)
5. Wide Area Network (WAN)



Types of Computer Networks

1. Personal Area Network (PAN)

PAN is the most basic type of computer network. This network is restrained to a single person, that is, communication between the computer devices is centered only on an individual's workspace. PAN offers a network range of 1 to 100 meters from person to device providing communication. Its transmission speed is very high with very easy maintenance and very low cost. This uses Bluetooth, IrDA, and Zigbee as technology. Examples of PAN are USB, computer, phone, tablet, printer, PDA, etc.



Personal Area Network (PAN)

Types of PAN

- **Wireless Personal Area Networks:** Wireless Personal Area Networks are created by simply utilising wireless technologies such as WiFi and Bluetooth. It is a low-range network.
- **Wired Personal Area Network:** A wired personal area network is constructed using a USB.

Advantages of PAN

- PAN is relatively flexible and provides high efficiency for short network ranges.
- It needs easy setup and relatively low cost.
- It does not require frequent installations and maintenance
- It is easy and portable.
- Needs fewer technical skills to use.

Disadvantages of PAN

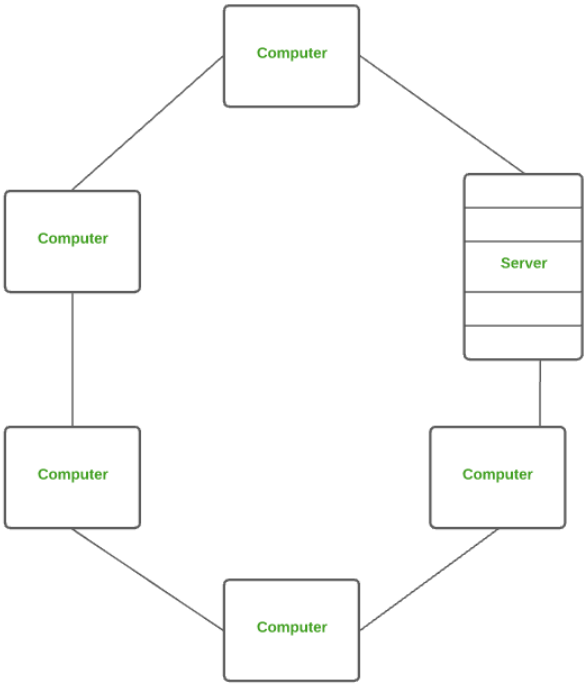
- Low network coverage area/range.
- Limited to relatively low data rates.
- Devices are not compatible with each other.
- Inbuilt WPAN devices are a little bit costly.

Applications of PAN

- Home and Offices
- Organizations and the Business sector
- Medical and Hospital
- School and College Education
- Military and Defense

2. Local Area Network (LAN)

LAN is the most frequently used network. A LAN is a computer network that connects computers through a common communication path, contained within a limited area, that is, locally. A LAN encompasses two or more computers connected over a server. The two important technologies involved in this network are Ethernet and Wi-fi. It ranges up to 2km & transmission speed is very high with easy maintenance and low cost. Examples of LAN are networking in a home, school, library, laboratory, college, office, etc.



Local Area Network (LAN)

Advantages of a LAN

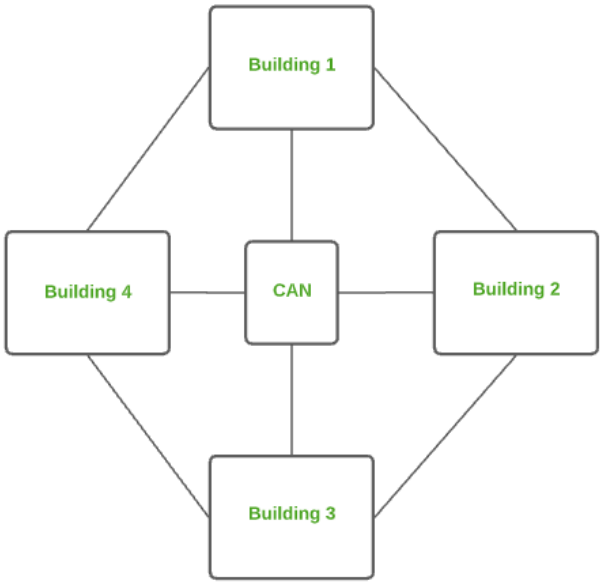
- **Privacy:** LAN is a private network, thus no outside regulatory body controls it, giving it a privacy.
- **High Speed:** LAN offers a much higher speed(around 100 mbps) and data transfer rate comparatively to WAN.
- **Supports different transmission mediums:** LAN support a variety of communications transmission medium such as an Ethernet cable (thin cable, thick cable, and twisted pair), fiber and wireless transmission.
- **Inexpensive and Simple:** A LAN usually has low cost, installation, expansion and maintenance and LAN installation is relatively easy to use, good scalability.

Disadvantages of LAN

- The initial setup costs of installing Local Area Networks is high because there is special software required to make a server.
- Communication devices like an ethernet cable, switches, hubs, routers, cables are costly.
- LAN administrator can see and check personal data files as well as Internet history of each and every LAN user. Hence, the privacy of the users are violated
- LANs are restricted in size and cover only a limited area
- Since all the data is stored in a single server computer, if it can be accessed by an unauthorized user, can cause a serious data security threat.

3. Campus Area Network (CAN)

CAN is bigger than a LAN but smaller than a MAN. This is a type of computer network that is usually used in places like a school or colleges. This network covers a limited geographical area that is, it spreads across several buildings within the campus. CAN mainly use Ethernet technology with a range from 1km to 5km. Its transmission speed is very high with a moderate maintenance cost and moderate cost. Examples of CAN are networks that cover schools, colleges, buildings, etc.



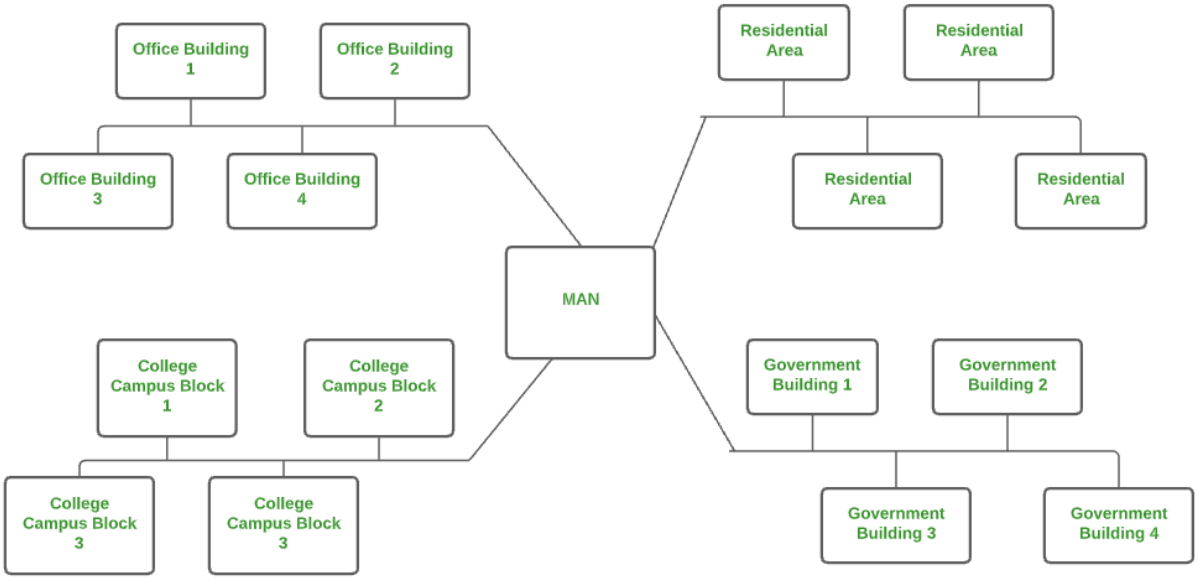
Campus Area Network (CAN)

Advantages of CAN

- **Speed:** Communication within a CAN takes place over Local Area Network (LAN) so data transfer rate between systems is little bit fast than Internet.
- **Security:** Network administrators of campus take care of network by continuous monitoring, tracking and limiting access. To protect network from unauthorized access firewall is placed between network and internet.
- **Cost effective:** With a little effort and maintenance, network works well by providing fast data transfer rate with multi-departmental network access. It can be enabled wirelessly, where wiring and cabling costs can be managed. So to work with in a campus using CAN is cost-effective in view of performance

4. Metropolitan Area Network (MAN)

A MAN is larger than a LAN but smaller than a WAN. This is the type of computer network that connects computers over a geographical distance through a shared communication path over a city, town, or metropolitan area. This network mainly uses FDDI, CDDI, and ATM as the technology with a range from 5km to 50km. Its transmission speed is average. It is difficult to maintain and it comes with a high cost. Examples of MAN are networking in towns, cities, a single large city, a large area within multiple buildings, etc.



Metropolitan Area Network (MAN)

Advantages of MAN

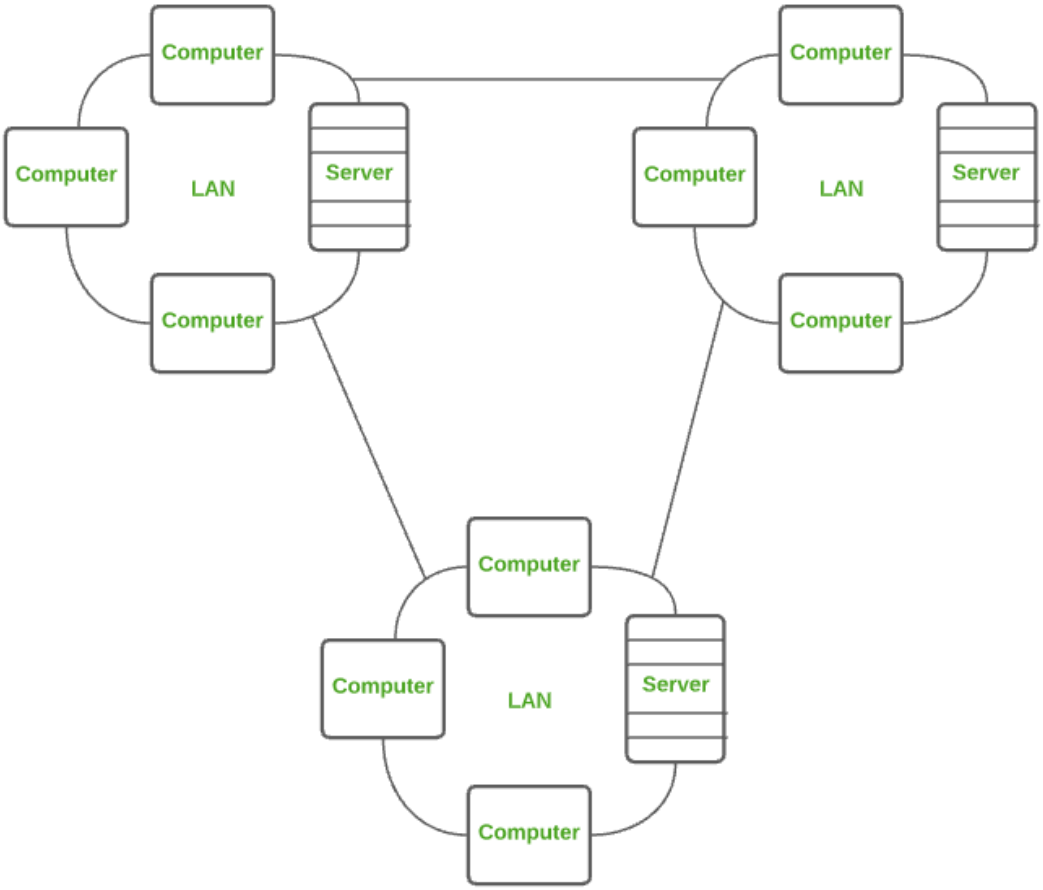
- MAN offers high-speed connectivity in which the speed ranges from 10-100 Mbps.
- The security level in MAN is high and strict as compared to WAN.
- It supports to transmit data in both directions concurrently because of dual bus architecture.
- MAN can serve multiple users at a time with the same high-speed internet to all the users.
- MAN allows for centralized management and control of the network, making it easier to monitor and manage network resources and security.

Disadvantages of MAN

- The architecture of MAN is quite complicated hence, it is hard to design and maintain.
- This network is highly expensive because it required the high cost to set up fiber optics.
- It provides less fault tolerance.
- The Data transfer rate in MAN is low when compare to LANs.

5. Wide Area Network (WAN)

WAN is a type of computer network that connects computers over a large geographical distance through a shared communication path. It is not restrained to a single location but extends over many locations. WAN can also be defined as a group of local area networks that communicate with each other with a range above 50km. Here we use Leased-Line & Dial-up technology. Its transmission speed is very low and it comes with very high maintenance and very high cost. The most common example of WAN is the Internet.



Wide Area Network (WAN)

Advantages of WAN

- It covers large geographical area which enhances the reach of organisation to transmit data quickly and cheaply.
- The data can be stored in centralised manner because of remote access to data provided by WAN.
- The travel charges that are needed to cover the geographical area of work can be minimised.
- WAN enables a user or organisation to connect with the world very easily and allows to exchange data and do business at global level.

Disadvantages of WAN

- Traffic congestion in Wide Area Network is very high.
- The fault tolerance ability of WAN is very less.
- Noise and error are present in large amount due to multiple connection point.
- The data transfer rate is slow in comparison to LAN because of large distances and high number of connected system within the network.

Comparison between Different Computer Networks

Parameters	PAN	LAN	CAN	MAN	WAN
Full Name	Personal Area Network	Local Area Network	Campus Area Network	Metropolitan Area Network	Wide Area Network
Technology	Bluetooth, IrDA, Zigbee	Ethernet & Wifi	Ethernet	FDDI, CDDi. ATM	Leased Line, Dial-Up
Range	1-100 m	Upto 2km	1 – 5 km	5-50 km	Above 50 km
Transmission Speed	Very High	Very High	High	Average	Low
Ownership	Private	Private	Private	Private or Public	Private or Public
Maintenance	Very Easy	Easy	Moderate	Difficult	Very Difficult
Cost	Very Low	Low	Moderate	High	Very High

- **What is an IP Address?**

All the computers of the world on the Internet network communicate with each other with underground or underwater cables or wirelessly. If I want to download a file from the internet or load a web page or literally do anything related to the internet, my computer must have an address so that other computers can find and locate mine in order to deliver that particular file or webpage that I am requesting. In technical terms, that address is called **IP Address or Internet Protocol Address**.

Let us understand it with another example, like if someone wants to send you a mail then he/she must have your home address. Similarly, your computer too needs an address so that other computers on the internet can communicate with each other without the confusion of delivering information to someone else's computer. And that is why each computer in this world has a unique IP Address. Or in other words, an IP address is a unique address that is used to identify computers or nodes on the internet. This address is just a string of numbers written in a certain format. It is generally expressed in a set of numbers for example 192.155.12.1. Here each number in the set is from 0 to 255 range. Or we can say that a full IP address ranges from 0.0.0.0 to 255.255.255.255. And these IP addresses are assigned by IANA(known as Internet Corporation For Internet Assigned Numbers Authority).

But what is Internet protocol? This is just a set of rules that makes the internet work. You are able to read this article because your computer or phone has a unique address where the page that you requested has been delivered successfully.

Working of IP addresses

The working of IP addresses is similar to other languages. It can also use some set of rules to send information. Using these protocols, we can easily send, and receive data or files to the connected devices. There are several steps behind the scenes. Let us look at them

- Your device directly requests your Internet Service Provider which then grants your device access to the web.
- And an IP Address is assigned to your device from the given range available.
- Your internet activity goes through your service provider, and they route it back to you, using your IP address.
- Your IP address can change. For example, turning your router on or off can change your IP Address.
- When you are out from your home location your home IP address doesn't accompany you. It changes as you change the network of your device.

Types of IP Address

IP Address is of two types:

1. IPv4: Internet Protocol version 4. It consists of 4 numbers separated by the dots. Each number can be from 0-255 in decimal numbers. But computers do not understand decimal numbers, they instead change them to binary numbers which are only 0 and 1. Therefore, in binary, this (0-255) range can be written as (00000000 – 11111111). Since each number N can be represented by a group of 8-digit binary digits. So, a whole IPv4 binary address can be represented by 32-bits of binary digits. In IPv4, a unique sequence of bits is assigned to a computer, so a total of (2^32) devices approximately = 4,294,967,296 can be assigned with IPv4.
IPv4 can be written as: *189.123.123.90*

Classes of IPv4 Address: There are around 4.3 billion IPv4 addresses and managing all those addresses without any scheme is next to impossible. Let's understand it with a simple example. If you have to find a word from a language dictionary, how long will it take? Usually, you will take less than 5 minutes to find that word. You are able to do this because words in the dictionary are organized in alphabetical order. If you have to find out the same word from a dictionary that doesn't use any sequence or order to organize the words, it will take an eternity to find the word. If a dictionary with one billion words without order can be so disastrous, then you can imagine the pain behind finding an address from 4.3 billion addresses. For easier management and assignment IP addresses are organized in numeric order and divided into the following 5 classes:

IP Class	Address Range	Maximum number of networks
Class A	1-126	126 (2^7-2)
Class B	128-191	16384
Class C	192-223	2097152
Class D	224-239	Reserve for multitasking
Class E	240-254	Reserved for Research and development

The 0.0.0.0 is a Non-routable address is that indicates an invalid, or inapplicable end-user address. A loopback address is a distinct reserved IP address range that starts from 127.0.0.0 ends at 127.255.255.255 though 127.255.255.255 is the broadcast address for 127.0.0.0/8. The loopback addresses are built into the IP domain system, enabling devices to transmit and receive the data packets. The loopback address 127.0.0.1 is generally known as localhost.

2. IPv6: But, there is a problem with the IPv4 address. With IPv4, we can connect only the above number of 4 billion devices uniquely, and apparently, there are much more devices in the world to be connected to the internet. So, gradually we are making our way to **IPv6 Address** which is a 128-bit IP address. In human-friendly form, IPv6 is written as a group of 8 hexadecimal numbers separated with colons(:). But in the computer-friendly form, it can be written as 128 bits of 0s and 1s. Since, a unique sequence of binary digits is given to computers, smartphones, and other devices to be connected to the internet. So, via IPv6 a total of (2^128) devices can be assigned with unique addresses which are actually more than enough for upcoming future generations.

IPv6 can be written as: 2011:0bd9:75c5:0000:0000:6b3e:0170:8394

Classification of IP Address

An IP address is classified into the following types:

1. Public IP Address: This address is available publicly and it is assigned by your network provider to your router, which further divides it to your devices. Public IP Addresses are of two types,

- **Dynamic IP Address:** When you connect a smartphone or computer to the internet, your Internet Service Provider provides you an IP Address from the range of available IP Addresses. Now, your device has an IP Address and you can simply connect your device to the Internet and send and receive data to and from your device. The very next time when you try to connect to the internet with the same device, your provider provides you with different IP Addresses to the same device and also from the same available range. Since IP Address keeps on changing every time when you connect to the internet, it is called a Dynamic IP Address.
- **Static IP Address:** Static address never changes. They serve as a permanent internet address. These are used by DNS servers. What are DNS servers? Actually, these are computers that help you to open a website on your computer. Static IP Address provides information such as device is located on which continent, which country, which city, and which Internet Service Provider provides internet connection to that particular device. Once, we know who is the ISP, we can trace the location of the device connected to the internet. Static IP Addresses provide less security than Dynamic IP Addresses because they are easier to track.

2. Private IP Address: This is an internal address of your device which are not routed to the internet and no exchange of data can take place between a private address and the internet.

3. Shared IP addresses: Many websites use shared IP addresses where the traffic is not huge and very much controllable, they decide to rent it to other similar websites so to make it cost-friendly. Several companies and email sending servers use the same IP address (within a single mail server) to cut down the cost so that they could save for the time the server is idle.

4. Dedicated IP addresses: A dedicated IP Address is an address used by a single company or an individual which gives them certain benefits using a private Secure Sockets Layer (SSL) certificate which is not in the case of a shared IP address. It allows to access the website or log in via File Transfer Protocol (FTP) by IP address instead of its domain name. It increases the performance of the website when the traffic is high. It also protects from a shared IP address that is black-listed due to spam.

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